

1. INTRODUCTION

1.1 Project Overview

Project Title: Flight Booking App (MERN Stack)

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This project is a full-stack web application that enables users to search, compare, and book flights online securely.

1.2 Purpose

The purpose of this project is to provide a secure, scalable, and user-friendly platform for flight booking using MERN stack.

2. IDEATION PHASE

2.1 Problem Statement

Users face difficulty in booking flights due to complex interfaces, hidden charges, and security concerns.

2.2 Empathy Map Canvas

User expects affordable tickets, secure payments, and transparent pricing.

2.3 Brainstorming

Modules identified: Authentication, Flight Search, Booking, Payment, Admin Dashboard.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stages: Entice → Enter → Engage → Exit → Extend.

3.2 Solution Requirement

Functional and Non-functional requirements defined including security, scalability, usability.

3.3 Data Flow Diagram

DFD Level 0 and Level 1 illustrate user interaction with system and database.

3.4 Technology Stack

Frontend: React.js | Backend: Node.js & Express.js | Database: MongoDB Atlas | Auth: JWT.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Solution addresses user pain points by simplifying booking and improving security.

4.2 Proposed Solution

A MERN-based web application providing secure flight search and booking features.

4.3 Solution Architecture

3-tier architecture: React Frontend → Express Backend → MongoDB Database with cloud deployment.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

2 Sprints executed with total 32 story points and velocity of 16 story points per sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Application tested for authentication, search performance, booking flow, and payment reliability.

7. RESULTS

7.1 Output Screenshots

Login Page, Registration Page, Flight Search Page, Booking Confirmation Page, Admin Dashboard.

8. ADVANTAGES & DISADVANTAGES

Advantages: Secure, scalable, user-friendly.

Disadvantages: Depends on internet connectivity and payment gateway integration.

9. CONCLUSION

The Flight Booking App successfully demonstrates full-stack development using MERN stack and fulfills business and technical requirements.

10. FUTURE SCOPE

Integration of real-time airline APIs, seat selection, loyalty rewards, and mobile application version.

11. APPENDIX

Source Code: GitHub Repository Link

Dataset Link: Not Applicable

GitHub & Project Demo Link: <https://github.com/GuttulaSwathi22/flight-booking-app-mern>